Categorical Compositional Distributional Semantics

Categorical Compositional Distributional Semantics (DisCoCat)

Two Traditions of Meaning: Formal and Distributional

The study of linguistic meaning has long been divided between two traditions with complementary strengths and weaknesses:

1. **Formal semantics**, following Montague [-@montague1973proper], assigns meaning compositionally: the meaning of a complex expression is a function of the meanings of its parts and the way they are syntactically combined. This tradition excels at capturing logical structure (quantification, negation, modality) but struggles with lexical meaning—it typically treats content words as unanalyzed primitives.

2. **Distributional semantics** assigns meaning empirically: a word's meaning is characterized by its distribution across contexts, typically encoded as a vector in a high-dimensional space. The distributional hypothesis—"You shall know a word by the company it keeps" [@firth1957papers]—traces to J.R. Firth's contextual theory of meaning and, independently, to Zellig Harris's [-@harris1954distributional] algebraic analysis of

From Static Embeddings to Contextual Representations: LLMs as Distributional Models

The distributional programme has undergone a dramatic computational intensification in the era of large language models (LLMs). The trajectory from classical co-occurrence matrices through static word embeddings to contextual transformers can be understood as a progressive *enrichment* of the distributional hypothesis itself:

1. **Static embeddings**: Mikolov et al.'s [-@mikolov2013efficient] Word2Vec (skip-gram and CBOW architectures) demonstrated that prediction-based training on local context windows produces word vectors exhibiting striking algebraic regularity—the famous "king − man + woman ≈ queen" analogy. Pennington et al.'s [-@pennington2014glove] GloVe incorporated global co-occurrence statistics via log-bilinear regression, yielding vectors whose inner products approximate pointwise mutual information. Both models instantiate the distributional hypothesis in its classical Firthian form:

The DisCoCat Compositional Framework

The Key Insight: Shared Compact Closed Structure

DisCoCat's central observation is that pregroup grammars and vector spaces share a common abstract structure: both are **compact closed categories**. This means there exists a *meaning functor*:

$$F : \mathbf{Preg} \to \mathbf{FVect} \tag{4.1}$$

from the pregroup grammar category (where objects are types and morphisms are grammatical reductions) to the category of finite-dimensional vector spaces (where objects are vector spaces and morphisms are linear maps).

Under this functor:

- Noun types n map to a vector space N (the noun space)
- Sentence types s map to a vector space S (the sentence space)
- A transitive verb of type $n^r \cdot s \cdot n^l$ maps to a tensor in $N \otimes S \otimes N$
- Pregroup contractions (cups/caps) map to the standard inner

Case Enrichment in the DisCoCat Framework

The present contribution lies in showing how case marking enriches the DisCoCat framework with explicit role structure. In a case-marked DisCoCat model:

1. **Typed noun spaces**: Instead of a single noun space N, we define case-specific spaces $N_{\text{NOM}}, N_{\text{ACC}}, N_{\text{DAT}}, \ldots$ Morphisms between these spaces encode case-changing operations (passivization, dative shift, etc.).

2. **Case-constrained composition**: The meaning functor F maps case-typed pregroup derivations to tensor contractions that respect case constraints. A verb seeking a nominative subject and accusative object contracts only with vectors from the appropriate spaces.

3. **Alignment as functor**: Cross-linguistic alignment differences correspond to different meaning functors. An accusative-language functor maps S and A arguments to the same noun space N_{NOM}; an ergative-language functor maps S and P to a shared N_{ABS}.

The Compact Closure Axiom and Diagrammatic Complexity Metrics

The Snake Equation (Zigzag Identity)

The algebraic engine of DisCoCat is the **compact closure** of the pregroup category: for every type n, the adjunction maps $\eta_n : 1 \to n \otimes n^r$ (cap) and $\varepsilon_n : n^r \otimes n \to 1$ (cup) satisfy the **snake equation** (also called the zigzag identity):

$$(\varepsilon_n \otimes 1_n) \circ (1_n \otimes \eta_n) = 1_n \tag{4.3}$$

In string-diagrammatic terms, a cup composed with a cap on adjacent wires "straightens out" into an identity wire—a zigzag that cancels into a straight line. This axiom is not merely a formal curiosity: it is the *engine* that makes pregroup type reductions work. Every grammatical contraction (noun canceling with verb argument) is an instance of the cup map ε, and every expansion (introducing an adjoint pair) is an instance of the cap map η. The snake equation guarantees that these contractions and expansions are well-behaved—they can be freely inserted and removed without

Extensions: From Sentence Meaning to Discourse Coherence

DisCoCirc: Distributional Compositional Circuits

The original DisCoCat framework operates at the sentence level: each sentence receives a vector meaning, but there is no mechanism for tracking how meanings interact across sentences. De Felice and Coecke [-@defelice2020discourse] address this with **DisCoCirc** (Distributional Compositional Circuits), which extends the categorical framework to handle discourse-level semantic structure.

DisCoCirc introduces *state wires* that persist across sentence boundaries, encoding the evolving states of discourse entities (characters, objects, topics). A sentence like "Alice chased Bob. He was terrified." is represented as a circuit where:

- Alice and Bob are wires that persist across both sentences
- The pronoun "He" is resolved by connecting its wire to Bob's wire
- The emotional state "terrified" updates the state information carried by Bob's wire